國立陽明交通大學
NATIONAL YANG MING CHIAO TUNG UNIVERSITY

Institute of Artificial Intelligence Innovation
Department of Computer Science

Embedded Operating System

HW04: USB Driver

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Purpose for this Homework

- **Part 1. Understand the original USB HID Mouse Driver**
  - Trace the original USB HID Mouse Driver
  - Write comment for each section or line of code in the original USB HID Mouse Driver
- **Part 2. Write your own USB driver for USB mouse**
  - After understand original implementation, write your own one
  - Add your own driver to menuconfig like HW03 & compile / install kernel
  - Prepare a second mouse & add it to VMware virtual machine.
  - Unbind the original driver & bind you own one to your second mouse.
  - Find your print message in dmesg after your own driver operating successfully.

Part 1. Understand the original UDB HID Mouse Driver

- **Trace the original USB HID mouse Driver**
  - The path of original USB HID mouse driver is -> ../drivers/hid/usbhid/usbmouse.c
  - write comments for each section of line of code in the original USB HID mouse driver

Part 2. Write your own USB Driver

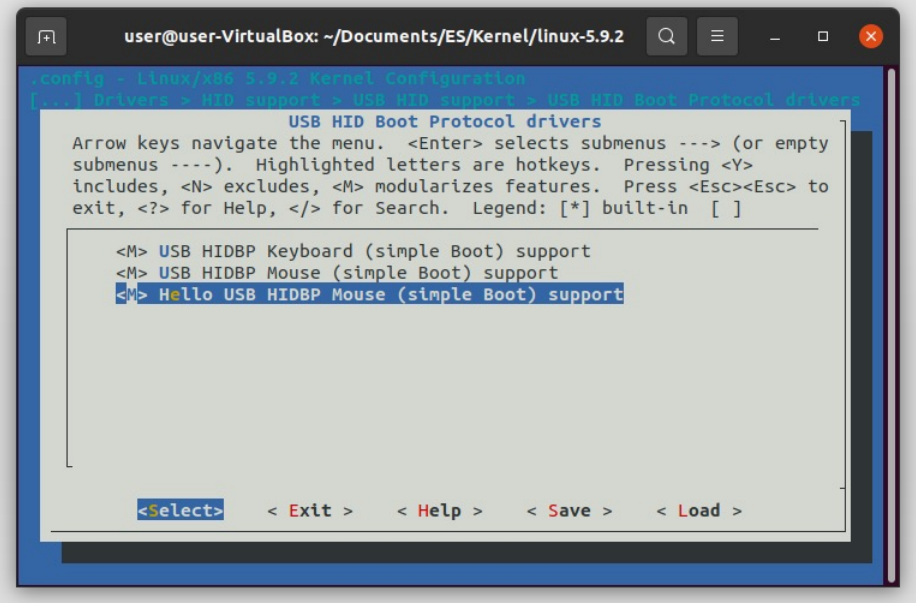
- **Write your own one**
  - **DO NOT COPY AND PASTE FROM** original implementation -> **0 point in this hw**
  - After writing your own one, leave comments as well
  - Add printk in your own driver for reporting left button click.
  - Remember to change the name in this struct as well.

```
static struct usb_driver usb_mouse_driver = {
    .name = "usbmouse_hello",
    .probe = usb_mouse_probe,
    .disconnect = usb_mouse_disconnect,
    .id_table = usb_mouse_id_table,
};
```

- **Add your own driver to kernel**
  - Add your own driver to menuconfig like HW03 & compile / install kernel
  - Alter ../drivers/hid/usbhid/Kconfig & Makefile like HW03
  - You will see something like this in menuconfig
  - Check the module you add as <M>
    - The path in menuconfig is

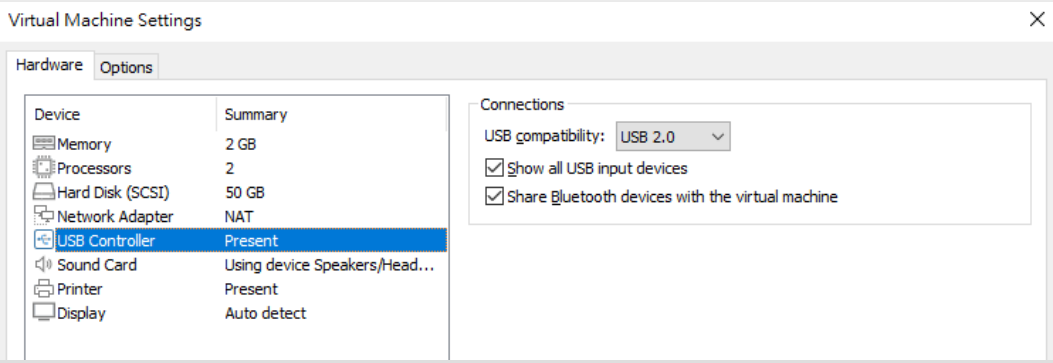
      Device Drivers -> HID support -> USB support

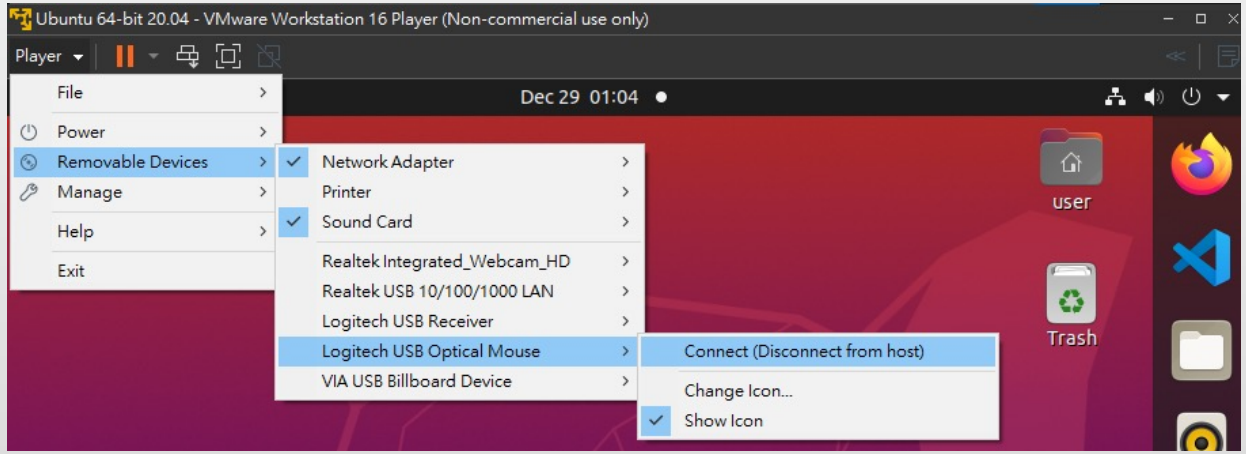
      -> USB HID Boot Protocal drivers
  - Use $make to recompile your kernel
  - After recompile your kernel, in you linux kernel source folder
    - sudo make modules_install
    - sudo make install



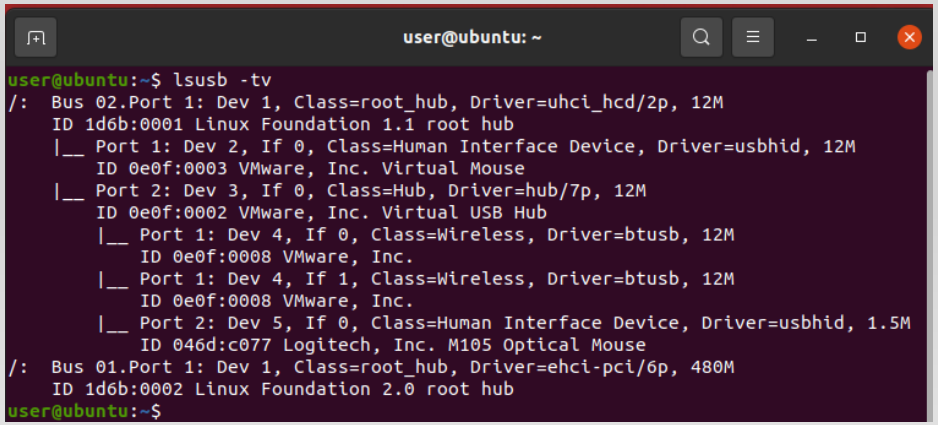
Part 2. Write your own USB Driver (cont'd)

- **Prepare a second mouse & add it to VMware virtual machine**
  - Completely shutdown(not suspend) the virtual machine.
  - Check the <Show all USB input devices> settings





  - Reboot the virtual machine, connect your second mouse.
  - Use $lsusb -tv to check your mouse and the driver it is binded with now.



Part 2. Write your own USB Driver (cont'd)

- **Unbind the original & bind you own one to your second mouse**
  - If you haven't set the password of root, use $sudo passwd root to set it. We will need it later.
  - Use $lsmod to check whether your driver is loaded or not. If it isn't, use $modprobe <module name> to load it.
  - Use $dmesg to find your second mouse's "usbid"
  - Use $su to change account to root.
  - $echo -n "usbid" > /sys/bus/usb/drivers/usbhid/unbind to unbind your second mouse with original driver.
  - $lsusb -tv to make sure you unbind second mouse driver successfully. Your second mouse should be unusable now.
  - $echo -n "usbid" > /sys/bus/usb/drivers/<your driver's name>/bind to bind your driver.
  - $lsusb -tv to make sure you bind your own driver successfully. Your second mouse should work now.

```
[ 2170.218732] Hello hid_generic_probe 109598085 !
[ 2170.218969] input: Logitech USB Optical Mouse as /devices/pci0000:00/0000:00:
11.0/0000:02:00.0/usb2/2-2/2-2.1/2-2.1:1.0/0003:046D:C077.0004/input/input8
[ 2170.219184] hid-generic 0003:046D:C077.0004: input,hidraw1: USB HID v1.11 Mou
se [Logitech USB Optical Mouse] on usb-0000:02:00.0-2.1/input0
```

```
user@user-VirtualBox:~$ lsusb -tv
/:  Bus 02.Port 1: Dev 1, Class=root_hub, Driver=uhci_hcd/2p, 12M
    ID 1d6b:0001 Linux Foundation 1.1 root hub
    |__ Port 1: Dev 2, If 0, Class=Human Interface Device, Driver=usbhid, 12M
        ID 0e0f:0003 VMware, Inc. Virtual Mouse
    |__ Port 2: Dev 3, If 0, Class=Hub, Driver=hub/7p, 12M
        ID 0e0f:0002 VMware, Inc. Virtual USB Hub
        |__ Port 1: Dev 6, If 0, Class=Human Interface Device, Driver=, 1.5M
            ID 046d:c077 Logitech, Inc. M105 Optical Mouse
/:  Bus 01.Port 1: Dev 1, Class=root_hub, Driver=ehci-pci/6p, 480M
    ID 1d6b:0002 Linux Foundation 2.0 root hub
user@user-VirtualBox:~$ lsusb -tv
/:  Bus 02.Port 1: Dev 1, Class=root_hub, Driver=uhci_hcd/2p, 12M
    ID 1d6b:0001 Linux Foundation 1.1 root hub
    |__ Port 1: Dev 2, If 0, Class=Human Interface Device, Driver=usbhid, 12M
        ID 0e0f:0003 VMware, Inc. Virtual Mouse
    |__ Port 2: Dev 3, If 0, Class=Hub, Driver=hub/7p, 12M
        ID 0e0f:0002 VMware, Inc. Virtual USB Hub
        |__ Port 1: Dev 6, If 0, Class=Human Interface Device, Driver=usbmouse_hello, 1.5M
            ID 046d:c077 Logitech, Inc. M105 Optical Mouse
/:  Bus 01.Port 1: Dev 1, Class=root_hub, Driver=ehci-pci/6p, 480M
    ID 1d6b:0002 Linux Foundation 2.0 root hub
user@user-VirtualBox:~$
```

```
root@user-VirtualBox:~# echo -n "2-2.1:1.0" > /sys/bus/usb/drivers/usbhid/unbind
root@user-VirtualBox:~# echo -n "2-2.1:1.0" > /sys/bus/usb/drivers/usbmouse_hello/bind
```

Requirement of this Homework

- **Screenshot of your source code**
  - The original USB HID driver with all your comments.
  - Your USB HID driver wrote with comments
- **Screenshot of USB & mouse information**
  - Your second mouse information before unbind original driver (lsusb -tv)
  - Your second mouse usbid (dmesg)
  - Your second mouse information after bind on your own driver (lsusb -tv)
- **Screenshot of left click log message**
  - left click your mouse 10 times, show your left click printk result (dmesg)

Submission

- Submit your PDF report in a zip format to the google Form.
- The google form link: https://forms.gle/VKXd6vTKkDTmib8v7
- The zip folder structure should look like this:
  - HW04_studentID.zip
    - report.pdf (your report with your screenshots)
    - USB_studentID_original.c (the original driver with your comments)
    - USB_studentID.c (your own driver with your comments)

Not like this:

- HW04_studentID.zip
  - folder
    - report.pdf (your report with your screenshots)
    - USB_studentID_original.c (the original driver with your comments)
    - USB_studentID.c (your own driver with your comments)

Grading

- Part 1. Understand the original USB HID Mouse Driver - 30%
- Part 2. Write your own USB driver for USB mouse - 70%
- If your report format or submission folder structure does not meet the requirements, 20% will be deducted from your total score. Please double-check before submitting.

Reminder

- In previous homework assignments 1 to 3, you should already be familiar with kernel modules and driver architecture.
- This Homework 4 will allow you to combine the knowledge you practiced before.
- This time, the TA will not help you with environment problems or code debugging; the TA will only assist with homework requirement issues.
- Please make good use of online information and do research by yourself.
- Any tools are allowed, but directly copying and pasting from the original driver or your classmates' implementations is not allowed and will result in a zero for this homework.

Q & A

Remember to submit your report.

The deadline is set as 6/12 23:59

Late submission will be not accepted!